An Experiment on the Effectiveness and Efficiency of Exploratory Testing DOI 10.1007/s10664-014-9301-4 Team members: Blezu Iosif | Bucur Mihnea | Bica Doru

Goal

To evaluate the effectiveness and efficiency of using the Bug Magnet browser extension during exploratory testing exercises conducted by software engineering students in a higher education setting.

Research Questions

1. Does the use of Bug Magnet improve the defect detection rate during exploratory testing tasks performed by students?
2. How does the use of Bug Magnet affect the time taken to identify critical issues?
3. What is the perceived usefulness and ease of use of Bug Magnet among student testers?

Methodology

- **Participants**: Undergraduate software engineering students enrolled in a software testing course.
- **Design**: A controlled experiment with two groups: a control group using standard exploratory testing, and an experimental group using Bug Magnet.
- **Task**: Each group will perform exploratory testing on the same web-based application for a fixed period.
- **Data Collection**: Logs of identified defects, timestamps, screen recordings, and post-session surveys.

Metrics

- **Effectiveness Metrics**:
    - Number of unique defects found
    - Severity distribution of identified defects
- **Efficiency Metrics**:
    - Average time to first defect detection
    - Total number of critical defects found per hour
- **Perception Metrics** (via post-session questionnaire):

- Perceived usefulness (Likert scale)
- Perceived ease of use (Likert scale)